

STUDIO 2 — SPATIAL DESIGN

Assessment Task — AT01

"The Narrow House That Breathes"

A Single Dwelling on an Infill Lot — Chippendale, Sydney

01 PROJECT OVERVIEW

Course	Spatial Design Studio 2
Assessment	AT01 — Design Project
Site	Chippendale, Inner Sydney NSW 2008
Site Dimensions	6m wide × 20m deep infill lot between two existing terraces
Program	Single dwelling — live / work / make (architect resident)
Scale	Approx. 120–150 sqm across 3 levels + roof terrace
Building Height	~9.4m floor to top slab / ~10m to parapet
Duration	2 weeks
Total Marks	100 marks
Submission Format	Digital PDF via Canvas + printed A2 sheets if required

02 DESIGN INTENT

This project asks you to design a home for a single architect — a person who lives, works, and makes within the same dwelling. The site is a narrow 6m wide infill lot in Chippendale, squeezed between two existing terrace houses. Your job is not to fight this constraint — it is to inhabit it.

| *How does a home carved between two walls find light, air, and identity?*

You are designing three zones stacked vertically — a ground floor studio/workspace, a first floor living area, and a second floor bedroom with access to a roof terrace. Each level has its own spatial character, its own relationship to light and privacy, and its own material presence. The section drawing is your primary design tool.

03 BUILDING HEIGHT & LEVELS

The building sits at approximately 9.4m from ground slab to top of second floor slab, with a parapet and roof terrace edge bringing the total to ~10m. This is consistent with standard 3-storey residential DCP height limits in Chippendale.

Level	Floor-to-Ceiling	Slab Thickness	Running Total
Ground Floor — Studio	3.2m	0.2m	3.4m
First Floor — Living	2.9m	0.2m	6.5m
Second Floor — Bedroom	2.7m	0.2m	9.4m
Parapet / Roof Terrace Edge	+0.6m	—	~10.0m total

Note: Ground floor has the tallest ceiling (3.2m) to give the workspace a generous, studio-like volume. The living floor is comfortable at 2.9m. The bedroom is most intimate at 2.7m. These height variations should read clearly in your section drawing.

04 EXTERIOR DESIGN LANGUAGE

The building has a split personality — a restrained, contextual street face and a generous, open rear. This duality is the exterior argument: public restraint, private openness. The building belongs to the street without mimicking its neighbours literally.

Street Facade — West-Facing (6m Wide)

– Material: Hand-laid dark brown brick — matching the tonal warmth of neighbouring Victorian terraces without copying their style

– One large pivoting steel-framed window at ground level — the studio visible from street, semi-public

– Narrow vertical slot windows on levels 1 and 2 — privacy maintained, rhythm echoes neighbouring terrace bays

– Recessed entry door set 600mm back from facade plane — creates a threshold moment without a porch

– No ornamentation — the brick bond pattern and mortar joint depth is the texture

– Roofline: flat parapet — clean contemporary top reads clearly against neighbouring Victorian hip and gable rooflines

Rear Facade — East-Facing (Opens to Garden)

- Full-width glazed sliding doors on first floor living level — morning north-east light floods the space
- Timber louvres on second floor bedroom — operable, warm contrast to brick front
- Small rear courtyard or garden — 3m deep, planted, provides green outlook from living level
- This face is where the building exhales — open, generous, soft

Design Logic: The dark brick exterior contextually belongs to Chippendale's Victorian terrace rhythm — same tonal family, same scale of facade bay — but is clearly of its own time. It does not try to replicate the decorative ironwork lacework or stucco detailing of its neighbours. It is a contemporary building that respects its context.

05 SITE RESEARCH — CHIPPENDALE STREETS

Before starting your Rhino model, you are required to conduct a visual site analysis of Chippendale using Google Maps Street View (or an in-person visit if possible). The following streets have been selected because they represent the most intact Victorian terrace fabric of the suburb and will directly inform your facade and site decisions.

Street	Character	What to Document
Abercrombie Street	Primary recommendation — most intact Victorian terrace rhythm in Chippendale	Brick colour and bond pattern, window proportions, fence setback from street, parapet heights, how new insertions sit next to heritage fabric
Shepherd Street	Tighter infill condition — newer buildings squeezed between old terraces	How contemporary insertions handle the party wall condition, setback vs alignment, materiality contrast between old and new
Balfour Street	Mixed residential grain, narrower street section	Entry threshold conditions, fence lines, letter box positions, how the building meets the footpath — very close relationship
Vine Street	Quiet residential, minimal traffic — best rear access views	Backyard conditions, rear-of-terrace relationships, lane access, planted vs paved rear zones
Rose Street	Southern edge of Chippendale, transition zone	Suburb edge character, scale change toward Camperdown, how building heights shift

Focus your research on Abercrombie Street first — it is the clearest example of the Victorian terrace typology your building will sit within. Document at minimum: the predominant brick tone (dark red-brown), the standard facade bay width (~4–6m), the typical parapet or hip roof profile, and the relationship of windows to solid wall.

06 PROGRAM BRIEF

Ground Floor — Studio / Workspace

- Open-plan work desk and making bench — tall ceiling creates studio volume
- Material storage wall along one party wall
- Direct street entry — semi-public threshold recessed 600mm
- Connection to internal light well or small rear courtyard

First Floor — Living

- Kitchen, dining, and living area as one open zone
- Full-width glazing to rear garden — morning light priority

- A reading alcove or secondary seating nook

Second Floor — Private / Rest

- Single bedroom with integrated storage
- Bathroom
- Access to roof terrace via a small hatch or stair

Circulation & Threshold

- The stair is not a leftover — it is a spatial event in the section
- Consider how the journey from street to bedroom is experienced level by level

07 DELIVERABLES

All drawings are to be produced in Rhino and composed in Adobe Illustrator or similar. Hand annotations and linework overlays are encouraged. All sheets are A2 portrait unless noted.

#	Deliverable	Specification	Marks
01	Site Analysis Sheet	1× A2 sheet. Chippendale block analysis — site dimensions, neighbouring built form, solar orientation (N arrow), street setbacks, and a 1:500 site plan. Include facade study sketches from street research. Hand-annotated diagrams encouraged.	10
02	Concept Diagram Sheet	1× A2 sheet. 3–4 diagrams communicating your design concept — spatial sequence, light strategy, privacy gradient from street to garden, and structural logic. No floor plans yet.	10
03	Floor Plans — All Levels	1× A2 sheet. Ground, first, and second floor plans at 1:50 drawn in Rhino. Include room labels, dimensions, north arrow, and furniture indication.	20
04	Longitudinal Section	1× A2 sheet. One long section cut through the full 20m depth at 1:50. Show floor levels (with annotated heights), ceiling heights, stair, roof, and light entry clearly. This is your primary drawing.	20
05	Cross Section + Section Perspective	1× A2 sheet. One cross section at 1:50 showing the full 6m width. Plus one section perspective (cut-away axonometric or perspective section) rendered to show atmosphere and materiality.	15
06	Interior Render — Two Views	1× A2 sheet. Two rendered interior perspectives from Rhino. One from the	15

#	Deliverable	Specification	Marks
		studio/ground floor, one from the living level. Show light, materiality, and spatial quality.	
07	Design Statement	300–400 words. Explain your concept, material choices, how your section responds to the narrow site, and how the exterior relates to Chippendale's terrace context.	10

Total: 100 marks

08 TWO-WEEK TIMELINE

Week	Days	Tasks	Milestone
1.1	Days 1–2	– Visit Abercrombie St + Shepherd St via Street View – Document brick tone, window proportions, parapet heights – Begin site analysis sheet and 1:500 site plan	Site analysis draft
1.2	Days 3–4	– Develop concept — sketch spatial sequence diagrams – Test 2–3 section ideas showing level changes and light strategy – Lock in your main concept direction	Concept diagrams draft
1.3	Days 5–7	– Begin Rhino model — site + party walls first – Develop floor plans at rough 1:100 – Test stair position and section relationship	Rhino model underway
2.1	Days 8–9	– Refine floor plans to 1:50 in Rhino – Produce longitudinal section — annotate all heights – Begin cross section	Plans + sections drafted
2.2	Days 10–11	– Set up interior render cameras in Rhino – Apply materials: dark brick exterior, timber floors, white plaster interior, steel stair – Render two interior views	Renders complete
2.3	Days 12–14	– Compose all 6 sheets in Illustrator or InDesign – Write 300–400 word design statement – Final PDF export and submission review	SUBMISSION DUE

09 ASSESSMENT CRITERIA

Criteria	Description	Marks
Conceptual Clarity	Does the project have a clear, legible design intent? Is there a spatial idea that runs consistently through all drawings?	20 marks
Section Quality	Is the longitudinal section the primary design driver? Does it reveal spatial thinking about height, light, level change, and the stair as event?	25 marks
Plan Logic	Do the three floor plans demonstrate clear spatial relationships, appropriate room sizing, and a considered circulation sequence?	20 marks
Render & Atmosphere	Do the two interior renders communicate light, materiality, and spatial quality? Is there a sense of how the house feels to inhabit?	20 marks
Presentation & Statement	Are sheets clearly composed, consistently scaled, and annotated? Does the design statement clearly explain the contextual and conceptual decisions?	15 marks

10 TECHNICAL NOTES & TIPS

Rhino — Getting Started

- Model the site first: draw the two party walls at 6m apart, 20m deep. These are your primary constraint.
- Layer structure from day one: SITE / PARTY-WALLS / STRUCTURE / FLOORS / WALLS / OPENINGS / FURNITURE / ROOF
- Use clipping planes for all section cuts — never manually draw section lines
- Set up named views early: STREET-ELEVATION / LONG-SECTION / CROSS-SECTION / CAM-GROUND / CAM-LIVING

Section Drawing

- Your longitudinal section cuts along the 20m depth — show every floor level, the stair, all ceiling heights, roof slab, and at least one skylight or high window
- Annotate heights on the section: 3.2m / 2.9m / 2.7m floors, slab thicknesses, overall 9.4m
- Lineweight hierarchy: 0.5mm cut elements, 0.25mm visible edges, 0.1mm background

Exterior Material Logic

- Brick: use a dark red-brown tone in your renders — reference Abercrombie Street's dominant brick colour
- Interior contrast: white plaster walls inside against the dark brick party walls creates a strong render image
- Stair: exposed steel stringer with timber treads — sits between the rawness of the brick and the warmth of timber floors

Renders

- Natural light only — no artificial lighting for this submission
- Morning light (north-east direction) for the living level render — this is when the rear glazing performs best
- Ground floor studio render should feel like a working space: cool, even light, slightly harder

Sheet Composition

- All A2 sheets: 594mm × 420mm
- Consistent title block: project title, sheet number, scale, your name, date
- White space is not wasted — margins and breathing room between drawings read as confidence

The section is not a byproduct of the plan. It is the design.

Architecture Studio 2 — University of Technology Sydney